BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Utility Software

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does operating-system file management do?

2. Why split network protocols into layers?

A) All layers must be identical
B) No interfaces are needed
C) Every packet becomes encrypted
D) Each layer has a job

3. "Encryption alone guarantees files cannot be stolen or deleted." Correct this. Explain why.

4. A fake useful app locks files for a ransom. "Malware is any app that uses lots of RAM." Correct this. How does this malware cause harm?

Today's objectives

Explain the core ideas of utility software using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Utility software

Carries out system maintenance or security tasks alongside the operating system.

Encryption utility

Protects stored data by transforming it into ciphertext.

Compression utility

Reduces file sizes for storage or transmission.

Defragmentation

Reorganises file fragments on spinning magnetic disks; not normally beneficial for SSDs.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Utility software	Carries out system maintenance or security tasks alongside the operating system.
Encryption utility	Protects stored data by transforming it into ciphertext.
Compression utility	Reduces file sizes for storage or transmission.
Defragmentation	Reorganises file fragments on spinning magnetic disks; not normally beneficial for SSDs.

I Do - One worked example

A school has a slow magnetic HDD with heavily fragmented files. Explain defragmentation.

1. Explain file fragments may be spread across disk positions.
2. Defragmentation rearranges related blocks contiguously.
3. This can reduce mechanical seek movement.
4. Avoid recommending it as a routine performance fix for SSDs.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A teacher must email a folder of large documents. Select a utility.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write the calculation for saving storage when a file is compressed.

```
originalMB = 500
compressedMB = 350
saved = ...
print(saved)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State a purpose of encryption utility software. [2 marks]

Q3. Explain why defragmentation can improve HDD access. [3 marks]

Q4. Explain why lossless compression matters for source code. [3 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Defragmenting an SSD routinely improves it in the same way as an HDD. ☐ T ☐ F
2. Carries out system maintenance or security tasks alongside the operating system. ☐ T ☐ F
3. Protects stored data by transforming it into ciphertext. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 It manages files and folders: naming, saving, copying, moving and deleting them. ☐

Q2 D - Each layer has a job ☐

Q3 Without the key, an attacker cannot easily read ciphertext. Encryption does not stop theft or deletion. ☐

Q4 Malware is made to harm, disrupt or gain access without permission. Different types work in different ways and can affect data access, privacy or availability. ☐

Code First - Model

```
originalMB = 500
compressedMB = 350
saved = originalMB - compressedMB
print(saved)
```

Output: 150

We Do - Check the model

1. Note lossless methods when exact files are required. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Converts readable data into ciphertext to restrict unauthorised reading. ☐

Q3 It places related fragments closer together, reducing mechanical seek time. ☐

Q4 Original bytes must be recovered exactly; lossy removal could corrupt instructions. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: